

Experiment 7:

Write a program that prints the addresses of code, global, static, heap, and stack variables. Analyze the Linux process address space layout and explain each memory segment.

Make use of the `/proc//maps` file and memory analysis tools to study the memory organization of a running process. Prepare a report explaining virtual memory mappings.

```
#include <stdio.h>
#include <stdlib.h>
#include <unistd.h>

int global_var = 10;
static int static_var = 20;

void code_function()
{
}

int main()
{
    int stack_var = 30;
    int *heap_var = malloc(sizeof(int));

    *heap_var = 40;

    printf("PID      : %d\n", getpid());
    printf("Code address : %p\n", (void *)code_function);
    printf("Global      : %p\n", (void *)&global_var);
    printf("Static      : %p\n", (void *)&static_var);
    printf("Heap        : %p\n", (void *)heap_var);
    printf("Stack       : %p\n", (void *)&stack_var);

    printf("\nPress Enter to exit...\n");
    getchar();

    free(heap_var);

    return 0;
}
```

```
gssggy@Gyanadeep:~$ cd ~
gssggy@Gyanadeep:~$ pwd
/home/gssggy
gssggy@Gyanadeep:~$ cd ~/OSLab
gssggy@Gyanadeep:~/OSLab$ pwd
/home/gssggy/OSLab
gssggy@Gyanadeep:~/OSLab$ mkdir -p Practical-07
gssggy@Gyanadeep:~/OSLab$ cd Practical-07
gssggy@Gyanadeep:~/OSLab/Practical-07$ nano practical7.c
gssggy@Gyanadeep:~/OSLab/Practical-07$ gcc practical7.c -o practical7
gssggy@Gyanadeep:~/OSLab/Practical-07$ ./practical7
PID          : 554
Code address : 0x5dffd292b209
Global       : 0x5dffd292e010
Static       : 0x5dffd292e014
Heap         : 0x5dffffd306010
Stack        : 0x7fff7bff59cc

Press Enter to exit...
|
```